Boa Jang

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RESEARCH SUMMARY

Medical AI researcher working with real-world clinical data to develop clinically translatable models. My work spans representation learning, biological aging assessment, and domain shift adaptation, with the broader goal of building AI systems that remain clinically meaningful and reliable beyond retrospective benchmark settings.

EDUCATION

Seoul National University — Seoul, Korea
Ph.D. Candidate in Bioengineering — Mar. 2023 – Present

Peking University — Beijing, China
B.S. in Psychological and Cognitive Sciences — Sep. 2016 – Jul. 2020

RESEARCH EXPERIENCE

Seoul National University Hospital (SNUH) — Seoul, Korea
Medical AI Researcher — Nov. 2021 – Feb. 2023
Ph.D. Research Collaboration with SNUH — Mar. 2023 – Present

Developed medical AI models from real-world clinical data in close collaboration with clinicians.

Korea Institute of Science and Technology (KIST) — Seoul, Korea
Research Intern, Brain Science Institute — Aug. 2020 – May 2021

Investigated neural circuits and motor control using in vivo behavioral experiments, optogenetics, and fluorescence imaging.

MANUSCRIPTS UNDER REVIEW / IN REVISION

[m.5] Retinal Age Gap as a Biomarker of Biological Aging Reflected by Socioeconomic Status and Lifestyle Behaviors. *GeroScience* (in revision, first author).

[m.4] Skeleton-Guided Progressive Test-Time Adaptation for Thin Curvilinear Structures. *AAAI 2027* (under review, first author).

[m.3] Integrating RNFL and Fundus Photographs for Differentiation of Glaucomatous and Non-Glaucomatous Optic Neuropathy Using Deep Learning. *Scientific Reports* (in revision).

[m.2] Prediction of Postoperative Intracerebral Hemorrhage Following Brain Biopsy Using Integrated Statistical and Machine Learning Analyses. *Scientific Reports* (in revision).

[m.1] Age- and Sex-Specific Autism Screening Using Machine Learning Reduction and Conversational Agent Integration. *Translational Psychiatry* (in revision).

PEER-REVIEWED PUBLICATIONS

† co-first author

[j.8] J.R. Song†, **B. Jang**†, et al. (2026). Segmentation-Guided Denoising Diffusion Probabilistic Models for Retinal OCT Speckle Noise Reduction with Enhanced Clinical Diagnostic Accuracy. *Scientific Reports* (Accepted).

[j.7] **B. Jang**, Y. Ahn, et al. (2026). Clinically Grounded Retinal Representation Learning from Minimal Supervision. *Scientific Reports.*

[j.6] **B. Jang**, Y. Shin, G. Lee, and Y.-G. Kim (2026). From Static Diagnosis to Dynamic Guidance: Evolution of Artificial Intelligence in Pediatric Neuroimaging. *Journal of Korean Neurosurgical Society.*

[j.5] J.H. Lee, S.Y. Lee, **B. Jang**, et al. (2026). OCT Biomarkers as Predictors of Treatment Interval in Neovascular Age-Related Macular Degeneration Treated with Intravitreal Aflibercept Using a Treat-and-Extend Regimen. *Scientific Reports.*

[j.4] **B. Jang**†, C.H. Lee†, et al. (2026). Artificial Intelligence-Based Prediction of First Recurrence in Neovascular Age-Related Macular Degeneration with Validation by 19 Experts. *Scientific Reports.*

[j.3] D.Y. Kim, R. Do, . . . , **B. Jang**, et al. (2025). Automated AI-Based Identification of Autism Spectrum Disorder from Home Videos. *npj Digital Medicine.*

[j.2] N.Y. Kim†, **B. Jang**†, et al. (2024). Differential Diagnosis of Pleural Effusion Using Machine Learning. *Annals of the American Thoracic Society.*

[j.1] **B. Jang**†, S.Y. Lee†, et al. (2023). Preliminary Analysis of Predicting the First Recurrence in Patients with Neovascular Age-Related Macular Degeneration Using Deep Learning. *BMC Ophthalmology.*

PRESENTATIONS

[o.7] **B. Jang**, G. Lee, J. Choi, and Y.-G. Kim. "Structure-Preserving Test-Time Adaptation for Cross-Modality Retinal Vessel Segmentation." *2026 Spring Conference of The Korean Society of Medical Informatics (KoSMI)*, Cheongju, Korea.

[po.3] **B. Jang**, K.H. Bae, and Y.-G. Kim. "Assessment of Systemic Health Using Retinal Age Gap: Development of a Predictive Model and Clinical Applicability." *2025 Annual Conference of the Korea Society of Artificial Intelligence in Medicine (KoSAIM)*, Korea.

[o.6] **B. Jang** et al. "Assessment of Systemic Health Using Retinal Age Gap: Development of a Predictive Model and Clinical Applicability." *MICCAI 2025 AMAI Workshop*, Daejeon, Korea. **Best Abstract Award – Honorable Mention**.

[o.5] **B. Jang** et al. "Assessment of Systemic Health Using Retinal Age Gap: Development of a Predictive Model and Clinical Applicability." *2025 Spring Conference of The Korean Society of Medical Informatics (KoSMI)*, Busan, Korea. **Best Paper Award**.

[o.4] **B. Jang** et al. "Self-Supervised Learning for Mesangial Proliferation Prediction in Digital Pathology." *The 45th Annual Meeting of the Korean Society of Nephrology (KSN 2025)*, Seoul, Korea.

[o.3] **B. Jang** and Y.-G. Kim. "Foundation Models in Healthcare: Development, Necessity and Implementation Strategies." *2025 HIMSS Global Health Conference & Exhibition (HIMSS 2025)*, Las Vegas, NV, USA.

[o.2] **B. Jang** et al. "The Impact of AI-Assistance on Predicting Recurrence in Neovascular Age-Related Macular Degeneration for Ophthalmologists." *2023 Fall Conference of The Korean Society of Medical Informatics (KoSMI)*, Bundang, Korea. **Best Paper Award**.

[po.2] **B. Jang** et al. "Development and Validation of Pretrained Model for Fundoscopy-Specific using Large-Scale Images." *10th Workshop on Ophthalmic Medical Image Analysis (OMIA-X), 2023 in International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, Vancouver, Canada.

[po.1] **B. Jang**, Y.-G. Kim, and E.K. Lee. "Predicting the First Recurrence in Patients with Neovascular AMD Using Deep Learning." *2023 American Society of Retina Specialists (ASRS)*, Seattle, WA, USA. **Poster Award Winner**.

[o.1] **B. Jang** et al. "Development and Validation of Pretrained Model for Fundoscopy-Specific using Large-Scale Images." *2023 Spring Conference of The Korean Society of Medical Informatics (KoSMI)*, Daegu, Korea. **Best Paper Award**.

AWARDS

[a.5]	AI Seoul Tech Graduate Scholarship, Seoul Future Talent Foundation	Nov. 2025
[a.4]	Hana Bank President's Award, 2024 Digital Wellness Competition	Nov. 2024
[a.3]	2nd Place, Healthcare AI Learning Platform Challenge 2023, Themes 4 & 6	Dec. 2023
[a.2]	3rd Place, Digital Pathology AI Hackathon, CODiPAI Challenge	Oct. 2023
[a.1]	SNUAIMED Excellence Scholarship	2023, 2024

TEACHING & MENTORING

[t.4]	Teaching Assistant, Korea Clinical Datathon (KCD) 2025	Oct. 2025
[t.3]	Student Ambassador, MICCAI 2025 AMAI Workshop	Sep. 2025
[t.2]	Teaching Assistant, Clinical and Healthcare AI Training Programs	2024, 2025
[t.1]	Teaching Assistant, NVIDIA MONAI Bootcamp, HCLS Summit Korea 2022	May 2022

PATENTS

[p.2] Y.-G. Kim, . . . , **B. Jang**. Controllable Normal-to-Disease Fundus Translation via Factorial Guidance. Korea Patent Application No. 10-2026-0119988, filed June 2026. *(Pending)*

[p.1] H.-S. Kim, . . . , Y.-G. Kim, **B. Jang**. Method and System for Predicting Information Related to Abdominal Surgery from Medical Image Data. Korea Patent No. 10-2953173, registered April 2026. *(Granted)*

REFERENCES

Prof. Jin-Wook Choi, PhD (jinchoi@snu.ac.kr)
Prof. Young-Gon Kim, PhD (younggon2.kim@gmail.com)
Prof. Hyuk Jin Choi, MD, PhD (docchoi@hanmail.net)